

Super-Exponential Light Transport: A Statistical Foundation

Part 1 — The Stochastic Extinction Field and Its Path Average

1.1 Setting up the problem

Classical radiative transfer treats the extinction coefficient $\sigma(\mathbf{x})$ as a deterministic, fully-resolved function of position. In practice, most real media contain structure at scales far below what any macroscopic model can resolve — individual particle positions, local density fluctuations, micro-scale clustering or exclusion. To account for this unresolved structure without explicitly modeling every particle, the extinction coefficient is instead written as the sum of a resolved mean and an unresolved, randomly-fluctuating part:

$$\sigma(\mathbf{x}) = \sigma_m + \sigma_\mu(\mathbf{x}) \quad (1)$$

Here σ_m is the macroscopic extinction coefficient — the value that would be measured or specified if the medium's fine structure were invisible — and $\sigma_\mu(\mathbf{x})$ is a random field representing everything left out of that macroscopic description. The field σ_μ is taken to have zero mean,

$$\langle \sigma_\mu(\mathbf{x}) \rangle = 0, \quad (2)$$

where $\langle \cdot \rangle$ denotes an ensemble average over all statistically possible realizations of the medium's micro-structure. This normalization is a deliberate modeling choice rather than an incidental simplification: it guarantees that σ_m alone already captures the correct macroscopic extinction, and that σ_μ can only redistribute density around that mean — concentrating it in some regions at the expense of others — without ever shifting the mean itself.

1.2 Optical depth as a random variable

In a medium described this way, the optical depth accumulated along a path is no longer a fixed function of distance. For a ray originating at $\mathbf{x}$ traveling in direction $\boldsymbol{\omega}$, the optical depth over a path length s is

$$\tau(\mathbf{x}, \boldsymbol{\omega}, s) = \int_0^s \sigma(\mathbf{x} + s' \boldsymbol{\omega}) ds'. \quad (3)$$

Because σ is a random field, τ is a random variable even for fixed $\mathbf{x}$, $\boldsymbol{\omega}$, and s : each realization of the medium's micro-structure produces a different value of τ for the same geometric path. This is the essential departure from classical transport, where optical depth is a deterministic function of distance alone.

1.3 The line-averaged extinction

It is convenient to work with the *line-averaged extinction* rather than the optical depth directly:

$$\bar{\sigma}_s = \frac{\tau(\mathbf{x}, \boldsymbol{\omega}, s)}{s} = \frac{1}{s} \int_0^s \sigma(\mathbf{x} + s' \boldsymbol{\omega}) ds'. \quad (4)$$

$\bar{\sigma}_s$ is the effective, constant extinction rate that a path of length s would need to have, under classical (non-random) transport, to accumulate the same optical depth actually encountered along that particular realization of the path. It is a random variable whose distribution is entirely determined by the statistics of σ_μ — specifically, as later parts will show, by how σ_μ correlates with itself at different points along the path.

1.4 The mean of the line-averaged extinction

The first and simplest statistical property of $\bar{\sigma}_s$ follows directly from the linearity of the ensemble average and the zero-mean assumption on σ_μ :

$$\langle \bar{\sigma}_s \rangle = \left\langle \frac{1}{s} \int_0^s \sigma(\mathbf{x} + s' \boldsymbol{\omega}) ds' \right\rangle = \frac{1}{s} \int_0^s \langle \sigma(\mathbf{x} + s' \boldsymbol{\omega}) \rangle ds' = \frac{1}{s} \int_0^s \sigma_m ds' = \sigma_m \quad (5)$$

This result holds for every path length s , regardless of how the fluctuations σ_μ are correlated or how large their variance is. It establishes the essential reference point for everything that follows: whatever the true distribution of $\bar{\sigma}_s$ turns out to be, its mean is always the macroscopic extinction coefficient, and any resulting model of transmittance must reduce exactly to the classical Beer–Lambert law in the limit where the fluctuations vanish.

What is *not* yet determined by this result is how $\bar{\sigma}_s$ is distributed *around* that mean — and it is this distribution, governed by the second moment (variance) and higher statistics of σ_μ , that determines whether a medium behaves classically, exhibits enhanced long-range transmission, or exhibits the sharply truncated, structurally uniform transport associated with spatial exclusion. That is the subject of Part 2.

Part 2 – Variance of the Line-Averaged Extinction

2.1 Reducing the variance to a double integral

Since σ_m in Equation (1) is a deterministic constant, it contributes nothing to the variance of $\bar{\sigma}_s$; only the random component σ_μ matters:

$$\text{Var}[\bar{\sigma}_s] = \frac{1}{s^2} \text{Var} \left[\int_0^s \sigma_\mu(\mathbf{x} + s'\boldsymbol{\omega}) ds' \right]. \quad (6)$$

Because $\langle \sigma_\mu \rangle = 0$ everywhere, the mean of the integral inside the variance is itself zero, so the variance reduces to a plain second moment:

$$\text{Var} \left[\int_0^s \sigma_\mu(\mathbf{x} + s'\boldsymbol{\omega}) ds' \right] = \left\langle \left(\int_0^s \sigma_\mu(\mathbf{x} + s'\boldsymbol{\omega}) ds' \right)^2 \right\rangle. \quad (7)$$

Expanding the square as a double integral and moving the ensemble average inside,

$$= \int_0^s \int_0^s \langle \sigma_\mu(\mathbf{x} + s'\boldsymbol{\omega}) \sigma_\mu(\mathbf{x} + s''\boldsymbol{\omega}) \rangle ds' ds''. \quad (8)$$

The quantity inside the average, $\langle \sigma_\mu(\mathbf{x}') \sigma_\mu(\mathbf{x}'') \rangle$, is by definition the **autocovariance function** of the field σ_μ evaluated at the two points $\mathbf{x}' = \mathbf{x} + s'\boldsymbol{\omega}$ and $\mathbf{x}'' = \mathbf{x} + s''\boldsymbol{\omega}$:

$$\text{cov}(\mathbf{x}', \mathbf{x}'') \equiv \langle \sigma_\mu(\mathbf{x}') \sigma_\mu(\mathbf{x}'') \rangle. \quad (9)$$

Combining (6)–(9), the variance of the line-averaged extinction is fully determined by the autocovariance of the underlying field:

$$\text{Var}[\bar{\sigma}_s] = \frac{1}{s^2} \int_0^s \int_0^s \text{cov}(\mathbf{x}', \mathbf{x}'') ds' ds''. \quad (10)$$

This is the central result linking a statistical description of the medium's micro-structure – its autocovariance – to an observable macroscopic quantity: the variability of the extinction actually experienced along paths of a given length.

2.2 Reduction under stationarity

Equation (10) is fully general but requires the covariance at every pair of points along the path. A substantial simplification is available if the field is assumed **statistically homogeneous (stationary)**: the covariance between two points then depends only on the separation between them, not on their absolute position,

$$\text{cov}(\mathbf{x}', \mathbf{x}'') = C(r), \quad r = |s' - s''|, \quad (11)$$

where $C(r)$ is the one-dimensional autocovariance function along the transect defined by the ray direction ω . Under this assumption, the double integral in (10) can be reduced to a single integral over the lag r . For any pair (s', s'') within $[0, s] \times [0, s]$, the set of pairs sharing a given separation r has measure $2(s - |r|)$; carrying out this standard reduction (using the symmetry $C(r) = C(-r)$, which holds for any real-valued stationary process) gives

$$\text{Var}[\bar{\sigma}_s] = \frac{2}{s^2} \int_0^s (s - r) C(r) dr. \quad (12)$$

Equation (12) is the working formula: given any model for the autocovariance $C(r)$ of the micro-scale field, the variance of the line-averaged extinction over a path of length s follows directly by quadrature.

2.3 A consistency check

Equation (12) should recover the point variance of the field itself in the limit of vanishingly short paths, since $\bar{\sigma}_s \rightarrow \sigma(\mathbf{x})$ as $s \rightarrow 0$. Expanding for small s , and assuming $C(r)$ is continuous at $r = 0$ so that $C(r) \approx C(0)$ over the (shrinking) domain of integration,

$$\text{Var}[\bar{\sigma}_s] \approx \frac{2}{s^2} C(0) \int_0^s (s - r) dr = \frac{2}{s^2} C(0) \cdot \frac{s^2}{2} = C(0), \quad s \rightarrow 0. \quad (13)$$

This is exactly the point variance of σ_μ , confirming that (12) behaves correctly in the appropriate limit.

2.4 What remains to be specified

Equation (12) makes explicit that everything about how a medium's micro-structure affects light transport — at least within the continuous-field description adopted here — is encoded entirely in the shape of $C(r)$. Two qualitatively distinct behaviors are possible depending on how quickly $C(r)$ falls off with separation:

- If $C(r)$ decays quickly enough to be **integrable** over all r (e.g. an exponential or otherwise short-range correlation), $\text{Var}[\bar{\sigma}_s]$ decreases as s grows, and the medium's fluctuations average out over long paths — approaching classical, uncorrelated behavior at large scales.
- If $C(r)$ decays slowly, or not at all, over the scales of interest — a signature of clustering, or **positive, long-range correlation** — $\text{Var}[\bar{\sigma}_s]$ remains large even for long paths, and the resulting transport departs strongly and persistently from the classical case.

A third possibility — $C(r)$ becoming negative at some separations, representing a tendency of the field to alternate rather than persist — is also admissible within Equation (12), and will be shown in Part 3 to *reduce* $\text{Var}[\bar{\sigma}_s]$ relative to a comparable positively-correlated model, bringing transport closer to the classical prediction. It will not, however, produce transport that falls below the classical prediction: that behavior, associated with genuine spatial exclusion, requires a different statistical description entirely, developed separately later.

Part 3 will make the relationship between $\text{Var}[\bar{\sigma}_s]$ and the resulting transmittance precise.

Part 3 — The Ensemble-Averaged Transmittance and the Direction of the Effect

3.1 From optical depth to transmittance

By definition (Section 1.3), the optical depth over a path of length s is exactly $\tau(\mathbf{x}, \boldsymbol{\omega}, s) = s \bar{\sigma}_s$. For a single, fixed realization of the medium's micro-structure, the transmittance is therefore the ordinary Beer–Lambert expression evaluated at that realization's own line-averaged extinction:

$$\text{Tr}_\mu(s) = e^{-s \bar{\sigma}_s}. \quad (14)$$

The physically observable transmittance — the quantity that would actually be measured, or that a rendering algorithm must reproduce — is the average of this expression over all statistically possible realizations of the medium:

$$\langle \text{Tr}(s) \rangle = \langle e^{-s \bar{\sigma}_s} \rangle. \quad (15)$$

This is the central object of interest. Nothing so far restricts how this average relates to the naive substitution of the *mean* extinction into the classical formula, $e^{-s \bar{\sigma}_m}$ — establishing that relationship is the purpose of this part.

3.2 Jensen's inequality

Jensen's inequality states that for any convex function φ and any random variable X with finite mean,

$$\langle \varphi(X) \rangle \geq \varphi(\langle X \rangle), \quad (16)$$

with equality if and only if X is almost surely constant (i.e. $\text{Var}[X] = 0$) or φ is affine.

The function of interest here is $\varphi(x) = e^{-sx}$ for fixed $s > 0$. Its second derivative,

$$\varphi''(x) = s^2 e^{-sx}, \quad (17)$$

is strictly positive for every real x , so φ is strictly convex everywhere. Jensen's inequality therefore applies unconditionally, with $X = \bar{\sigma}_s$:

$$\langle e^{-s\bar{\sigma}_s} \rangle \geq e^{-s\langle \bar{\sigma}_s \rangle}. \quad (18)$$

Substituting the Part 1 result $\langle \bar{\sigma}_s \rangle = \sigma_m$ (Equation 5) gives the key inequality:

$$\boxed{\langle \text{Tr}(s) \rangle \geq e^{-s\sigma_m}} \quad (19)$$

with equality if and only if $\text{Var}[\bar{\sigma}_s] = 0$.

3.3 An unconditional result

Equation (19) holds regardless of the shape or sign of the autocovariance $C(r)$ introduced in Part 2. The only property of the fluctuating field that enters the argument is that $\text{Var}[\bar{\sigma}_s]$, as given by Equation (12), is a variance – and a variance is, by construction, never negative. Consequently:

Any medium whose extinction rate fluctuates around a mean value – however that fluctuation is spatially structured – transmits, on average, *at least as much* light as a homogeneous medium with the same mean extinction, and strictly more whenever the fluctuation has any variance at all.

This is a stronger and more general statement than it may first appear. It does not merely say that *clustering* increases transmittance; it says that within this modeling framework, **no choice of spatial correlation can decrease transmittance below the classical prediction**. A covariance function with negative lobes – representing, within a continuous random field,

some tendency of the fluctuation to alternate rather than persist – can make $\text{Var}[\bar{\sigma}_s]$ smaller than a strongly clustered alternative would produce, bringing the transmittance closer to the classical value. It cannot, however, drive $\text{Var}[\bar{\sigma}_s]$ negative, and therefore cannot push $\langle \text{Tr}(s) \rangle$ below $e^{-s\sigma_m}$.

3.4 The boundary of the framework

This conclusion is a direct consequence of the specific construction adopted in Parts 1–2: the extinction coefficient is modeled as a continuous random field, and transmittance is obtained by averaging the classical exponential over realizations of that field. This is precisely the modeling assumption that makes Equation (14) – a single, well-defined optical depth per realization – meaningful in the first place.

It follows that media exhibiting the opposite behavior – transmittance that falls *below* the classical prediction, decaying faster than any homogeneous medium with the same mean extinction could – cannot be produced by any random field constructed this way, regardless of how its autocovariance is chosen. Such behavior requires a statistical description of a fundamentally different kind: one in which the randomness resides in the discrete positions of scattering centers themselves, subject to a mutual exclusion constraint, rather than in a continuous local rate. The relevant mathematical object in that case is a point process with a hard or soft minimum-separation constraint, not a covariance function of a continuous field, and it falls outside the scope of the argument developed here.

Part 4 will develop the statistical foundation needed for that alternative description. Specific covariance models for the continuous-field case, and the closed-form transmittances permitted by Equation (19), remain for a later part.

Part 4 – Foundations of the Exclusion Framework: Point Processes and Renewal Theory

4.1 A different kind of randomness

The framework developed in Parts 1–3 places the randomness in a continuous field $\sigma(\mathbf{x})$ and asks how averaging a deterministic transport law over that field's realizations affects the observable transmittance. Media governed by spatial exclusion require a different starting point: the randomness resides not in a continuous rate, but in the positions of discrete scattering centers themselves, subject to the constraint that no two centers may lie closer together than some minimum separation.

Consider the positions of scattering centers along a single ray, ordered as $0 < x_1 < x_2 < x_3 < \dots$, and define the sequence of inter-particle spacings

$$\Delta_i = x_i - x_{i-1}, \quad x_0 \equiv 0. \quad (20)$$

If the Δ_i are independent and identically distributed, this is a **renewal process**, and the common distribution of Δ_i – denoted $p_c(s)$, with survival function $\text{Tr}_c(s) = \int_s^\infty p_c(s') ds'$ and mean $\mu = \int_0^\infty s p_c(s) ds$ – completely characterizes the point process.

The classical, uncorrelated case is the special instance in which Δ_i is exponentially distributed with rate σ_i : this is precisely the defining property of a Poisson process, and it is the reason classical transport has an exponential free-flight law in the first place. Spatial exclusion is introduced by decomposing each gap as

$$\Delta_i = \hat{s} + Y_i, \quad (21)$$

where $\hat{s} \geq 0$ is a fixed minimum separation enforced between every pair of consecutive centers, and $Y_i \geq 0$ is an independent, identically distributed residual spacing beyond that floor. The classical case is recovered exactly when $\hat{s} = 0$ and Y_i is exponential.

4.2 Two distinct free-flight distributions

A photon entering the medium does not, in general, begin its path at a particle position – it begins at an arbitrary point along the ray, uncorrelated with the renewal process governing particle placement. The distance from such an arbitrary starting point to the *first* scattering center it encounters is therefore not the same random variable as the gap Δ_i between two consecutive particles, and in general does not follow the same distribution. Following the terminology already introduced for p_c , this second distribution is denoted $p_u(s)$: the free-flight distribution from an *uncorrelated* (particle-independent) origin, as opposed to $p_c(s)$, the free-flight distribution between two *correlated* (consecutive-particle) origins.

The relationship between the two is a standard result of renewal theory, derived here from first principles. Consider observing the process over a long stretch $[0, T]$ containing many gaps, and select a point $t \in [0, T]$ uniformly at random, representing an arbitrary photon entry point. The probability that t falls within a particular gap is proportional to the *length* of that gap – a longer gap occupies more of the interval $[0, T]$ and is correspondingly more likely to contain the randomly chosen point. This length-biasing gives the density of the gap length *containing* a randomly selected point as

$$f_{\text{biased}}(x) = \frac{x p_c(x)}{\mu}, \quad (22)$$

normalized by μ so that f_{biased} integrates to one. Given that the selected point lies within a gap of length x , its position within that gap is uniformly distributed (by the symmetry of a uniformly random selection over $[0, T]$), so the distance from the selected point to the *end* of its containing gap — i.e., to the next particle — is uniform on $[0, x]$. The distribution of this residual distance, marginalized over the length-biased gap distribution, is

$$p_u(s) = \int_s^\infty \frac{1}{x} f_{\text{biased}}(x) dx = \int_s^\infty \frac{p_c(x)}{\mu} dx = \frac{\text{Tr}_c(s)}{\mu}. \quad (23)$$

This is the central relationship of this section:

$$\boxed{p_u(s) = \frac{\text{Tr}_c(s)}{\mu}} \quad (24)$$

The free-flight distribution seen from an arbitrary entry point is the *survival function* of the particle-to-particle gap distribution, normalized by its mean — not the gap distribution itself. In general, $p_u \neq p_c$.

4.3 Consistency check: the classical limit

For the classical case, $\Delta_i \sim \text{Exponential}(\sigma_t)$, so $p_c(s) = \sigma_t e^{-\sigma_t s}$, $\text{Tr}_c(s) = e^{-\sigma_t s}$, and $\mu = 1/\sigma_t$. Substituting into Equation (24),

$$p_u(s) = \frac{e^{-\sigma_t s}}{1/\sigma_t} = \sigma_t e^{-\sigma_t s} = p_c(s). \quad (25)$$

The two distributions coincide exactly. This is the point-process expression of memorylessness: for a Poisson process, the distance to the next event is statistically identical whether measured from a particle or from an arbitrary point, which is precisely why classical transport theory never needs to distinguish between the two. Equation (24) shows that this coincidence is a special property of the exponential distribution specifically, not a general feature of renewal processes — for any non-exponential p_c , the two distributions genuinely differ, and a model built on discrete, correlated scattering centers must track both.

With this distinction established, the next step is to introduce the minimum-separation decomposition of Equation (21) explicitly into p_c and derive the resulting closed-form transmittance — the mechanism by which a hard exclusion constraint produces decay faster than the classical exponential.

Part 5 – The Minimum-Separation Model and Its Transmittance

5.1 Specifying the residual spacing

Part 4 introduced the decomposition $\Delta = \hat{s} + Y$ without specifying the distribution of the residual spacing Y . The simplest choice consistent with the exclusion floor is to assume that, once the hard minimum separation has been satisfied, the remaining spacing is otherwise memoryless: $Y \sim \text{Exponential}(\lambda)$, with density $\lambda e^{-\lambda y}$ for $y \geq 0$. This isolates the effect of the exclusion constraint itself, without introducing any additional correlation structure beyond it.

Under this choice, the particle-to-particle gap density follows directly:

$$p_c(s) = \begin{cases} 0 & 0 \leq s < \hat{s} \\ \lambda e^{-\lambda(s-\hat{s})} & s \geq \hat{s} \end{cases}, \quad \text{Tr}_c(s) = \begin{cases} 1 & 0 \leq s < \hat{s} \\ e^{-\lambda(s-\hat{s})} & s \geq \hat{s} \end{cases}. \quad (26)$$

The mean gap is $\mu = \hat{s} + \mathbb{E}[Y] = \hat{s} + 1/\lambda$.

5.2 The free-flight distribution from an arbitrary origin

Applying the renewal-theory relationship derived in Part 4 (Equation 24), $p_u(s) = \text{Tr}_c(s)/\mu$, together with $\text{Tr}_u(s) = \int_s^\infty p_u(s') ds'$, gives the free-flight distribution seen from an entry point uncorrelated with the particle positions:

$$p_u(s) = \begin{cases} \frac{1}{\mu} & 0 \leq s < \hat{s} \\ \frac{e^{-\lambda(s-\hat{s})}}{\mu} & s \geq \hat{s} \end{cases} \quad (27)$$

Integrating directly and using $\mu = \hat{s} + 1/\lambda$ to simplify,

$$\text{Tr}_u(s) = \begin{cases} 1 - \frac{s}{\mu} & 0 \leq s < \hat{s} \\ \frac{e^{-\lambda(s-\hat{s})}}{\lambda\mu} & s \geq \hat{s} \end{cases} \quad (28)$$

Both branches agree at $s = \hat{s}$ (each equals $1/(\lambda\mu)$), confirming Tr_u is continuous, as any valid survival function must be.

The qualitative shape of Equation (28) already reveals the mechanism at work: for a fresh entry point, the survival probability decreases *linearly*, not exponentially, over the exclusion zone $[0, \hat{s})$ – reflecting the uniform probability of landing at any position within the gap enclosing the entry point – before switching to ordinary exponential decay beyond $\hat{s}$. There is no probability, at any entry point, of an immediate collision within the excluded region; the entire distribution of first-collision distances is pushed outward relative to the classical case.

5.3 Confirming decay faster than the classical case

The classical (Poisson) point process with the same mean spacing μ has gap distribution $\text{Exponential}(1/\mu)$, with survival function $e^{-s/\mu}$. A general, mean-independent measure of how far a distribution departs from this classical reference is its **coefficient of variation**,

$$\text{CV}^2 = \frac{\text{Var}[\Delta]}{\mu^2}, \quad (29)$$

which equals exactly 1 for any exponential distribution, and serves as the dividing line between "burstier than classical" ($\text{CV} > 1$, associated with clustering) and "more regular than classical" ($\text{CV} < 1$, associated with exclusion). For the minimum-separation model, since $\hat{s}$ is a fixed constant, $\text{Var}[\Delta] = \text{Var}[Y] = 1/\lambda^2 = (\mu - \hat{s})^2$, giving

$$\text{CV}^2 = \left(1 - \frac{\hat{s}}{\mu}\right)^2 < 1 \quad \text{whenever } \hat{s} > 0. \quad (30)$$

This confirms, in general and independent of any particular value of s , that the minimum-separation model produces spacings that are strictly more regular than the classical case for any nonzero exclusion floor.

The effect on transmittance itself can be seen directly at $s = \hat{s}$: the exclusion model gives $\text{Tr}_u(\hat{s}) = 1 - \hat{s}/\mu$ (Equation 28), while the classical reference with the same mean gives $e^{-\hat{s}/\mu}$. The standard inequality $e^{-x} > 1 - x$, valid for every $x > 0$ (a consequence of the same convexity property used in Part 3), shows immediately that

$$e^{-\hat{s}/\mu} > 1 - \frac{\hat{s}}{\mu} = \text{Tr}_u(\hat{s}), \quad (31)$$

so the exclusion model's survival probability at $s = \hat{s}$ is strictly *below* the classical prediction with the same mean spacing. Together with Equation (30), this confirms that the minimum-separation model does what Part 3 showed a continuous random field never can: it produces transmittance that falls below, not above, the classical Beer–Lambert law.

5.4 The classical limit

As $\hat{s} \rightarrow 0$, the exclusion floor vanishes and $\mu \rightarrow 1/\lambda$. Equation (26) reduces to $p_c(s) = \lambda e^{-\lambda s}$ for all $s \geq 0$ – an ordinary exponential distribution – and Equation (27) reduces to the identical expression for $p_u(s)$, exactly as the memorylessness argument of Part 4 requires. The minimum-separation model contains classical, uncorrelated transport as the special case in which the exclusion constraint is switched off entirely.

Part 6 – Synthesis: Two Frameworks, One Underlying Question

6.1 A shared reference point

Both frameworks developed in this document address the same underlying question – how does correlation among scattering centers alter light transport relative to the classical, uncorrelated case? – using genuinely different mathematical objects. Parts 1–3 describe the medium as a continuous, randomly fluctuating extinction field and study the transmittance obtained by averaging over its realizations. Parts 4–5 describe the medium as a discrete point process and study the distribution of gaps between consecutive scattering centers directly. Despite this difference in construction, both frameworks share a single reference point: the classical, memoryless (Poisson) medium, recovered exactly as a limiting or degenerate case in each – as $\text{Var}[\bar{\sigma}_s] \rightarrow 0$ in the equality condition of Part 3's Jensen's-inequality bound, and as $\hat{s} \rightarrow 0$ in Part 5.

This shared baseline suggests a common, dimensionless language for comparing the two: the coefficient of variation of the relevant free-flight distribution, $CV^2 = \text{Var}[\Delta]/\mu^2$, which equals exactly 1 for the exponential distribution underlying classical transport, and departs from 1 in a direction and magnitude that reflects how strongly, and in which sense, a given medium deviates from that baseline.

6.2 An essential asymmetry

What has actually been proven in this document, however, is not symmetric between the two frameworks, and this asymmetry is the central finding of the analysis as a whole.

Part 3 established, via Jensen's inequality, that a continuous randomly-fluctuating field can only ever produce ensemble-averaged transmittance *at or above* the classical prediction – a conclusion that holds regardless of the specific autocovariance structure chosen, and fails to hold only in the trivial case of zero variance. The continuous-field framework, in other words, is capable of representing correlation that increases variability in the free-flight distribution above the classical baseline, but is structurally incapable of representing the opposite.

Part 5 showed, by explicit construction, that a point process with a hard minimum-separation constraint produces exactly the behavior the continuous-field framework cannot reach: transmittance strictly *below* the classical prediction, with a coefficient of variation strictly less than one. Nothing in the point-process construction of Part 4, however, restricts it to this regime — a renewal process with a sufficiently heavy-tailed residual spacing Y , rather than the exponential residual assumed in Part 5, would produce $CV > 1$ and clustering behavior of the same qualitative character addressed by the continuous-field framework, through an entirely different mechanism.

The point-process description is therefore the more general of the two: it is capable, in principle, of representing both directions of departure from the classical baseline, distinguished only by the shape of a single gap distribution. The continuous-field, ensemble-averaging description is best understood not as an alternative of equal scope, but as a convenient and physically natural approximation for one half of that range — the half in which describing the medium through a smoothly fluctuating density is more tractable than describing it through explicit particle statistics — with Part 3 giving the precise statement of which half that is.

6.3 Closing remarks

Framed this way, the distinction between clustering and exclusion is not a distinction between two separate physical phenomena requiring unrelated theories, but a single question — how does the free-flight distribution depart from the memoryless exponential? — answered by a single parameter, the coefficient of variation, with a well-defined and asymmetric mapping to the mathematical machinery available to describe each side. Media that cluster admit a convenient description in terms of continuous density fluctuations, for which Jensen's inequality guarantees the direction of the resulting departure from classical transport before any specific model is even chosen. Media that exclude admit no such continuous-field description, and require instead the explicit statistics of a point process — but reward that additional structure with access to a regime the continuous-field approach cannot reach at all.

Part 7 — A Closed-Form Example for the Clustering Case

Part 5 produced a fully explicit transmittance model for the exclusion side of the classification. The continuous-field side, by contrast, was so far characterized only by the general bound of Equation (19): a statement about the *direction* of the departure from classical transport, not its specific magnitude. This part closes that gap with an explicit, worked example.

7.1 From variance to a closed-form transmittance

Equation (19) constrains $\langle \text{Tr}(s) \rangle$ but does not determine it: the exact value of $\langle e^{-s\bar{\sigma}_s} \rangle$ depends on the full distribution of $\bar{\sigma}_s$, not merely its mean and variance. Obtaining a closed form therefore requires committing to a specific parametric family for that distribution. Since $\bar{\sigma}_s$ represents a physical extinction rate, any admissible family must be supported only on non-negative values – ruling out the Gaussian distribution, whose support extends to $-\infty$, in favor of the **Gamma distribution**, which is non-negative by construction and fully determined by a mean and a variance.

Matching a Gamma distribution's mean and variance to the results already established – $\langle \bar{\sigma}_s \rangle = \sigma_m$ (Equation 5) and $\text{Var}[\bar{\sigma}_s] = V(s)$ (Equation 12, for whichever $C(r)$ is chosen) – fixes its shape and scale parameters by the standard method of moments:

$$k = \frac{\sigma_m^2}{V(s)}, \quad \theta = \frac{V(s)}{\sigma_m}. \quad (32)$$

The moment generating function of a $\text{Gamma}(k, \theta)$ random variable is $M(t) = (1 - \theta t)^{-k}$ for $t < 1/\theta$, so the ensemble-averaged transmittance is obtained immediately by evaluating at $t = -s$:

$$\langle \text{Tr}(s) \rangle = \left(1 + \frac{s V(s)}{\sigma_m} \right)^{-\sigma_m^2/V(s)}. \quad (33)$$

Consistency check. As $V(s) \rightarrow 0$, Equation (33) must reduce to the classical Beer–Lambert law. Writing $\varepsilon = V(s)$, $a = s/\sigma_m$, $b = \sigma_m^2$, Equation (33) has the form $(1 + a\varepsilon)^{-b/\varepsilon}$, and the standard limit $(1 + a\varepsilon)^{1/\varepsilon} \rightarrow e^a$ as $\varepsilon \rightarrow 0$ gives

$$\lim_{V(s) \rightarrow 0} \langle \text{Tr}(s) \rangle = e^{-ab} = e^{-s\sigma_m}, \quad (34)$$

confirming the correct classical limit. It can also be verified directly, for any $V(s) > 0$, that Equation (33) satisfies the bound of Equation (19) – as it must, being the exact transmittance of a genuine, non-negative random variable with mean σ_m , to which Jensen's inequality applies by construction.

7.2 A specific covariance model: short-range correlation

Equation (33) is general, applying to any autocovariance $C(r)$ through $V(s)$. A concrete example is obtained by choosing an exponentially decaying covariance,

$$C(r) = v e^{-r/\xi}, \quad (35)$$

with $v = C(0)$ the point variance of σ_μ and ξ a characteristic correlation length – the simplest representative of the short-range, integrable class of covariance functions identified in Section 2.4. Substituting Equation (35) into Equation (12) and evaluating the integral gives

$$V(s) = \frac{2v\xi}{s^2} \left(s - \xi + \xi e^{-s/\xi} \right). \quad (36)$$

Two checks confirm this expression before it is used further. As $s \rightarrow 0$, expanding $e^{-s/\xi}$ to second order gives $V(s) \rightarrow v$, recovering the point variance exactly, as required by the general result of Equation (13). As $s \rightarrow \infty$, $V(s) \rightarrow 2v\xi/s \rightarrow 0$: the fluctuations average out over paths much longer than the correlation length, exactly the short-range behavior anticipated in Section 2.4.

7.3 The resulting transmittance

Substituting Equation (36) into Equation (33) gives a fully explicit, closed-form transmittance model for a medium with short-range, exponentially correlated micro-scale fluctuations:

$$\langle \text{Tr}(s) \rangle = \left(1 + \frac{2v\xi(s - \xi + \xi e^{-s/\xi})}{\sigma_m s} \right)^{-\frac{\sigma_m^2 s^2}{2v\xi(s - \xi + \xi e^{-s/\xi})}}. \quad (37)$$

Equation (37) depends on three physically interpretable parameters – the mean extinction σ_m , the point variance v of the micro-scale fluctuation, and the correlation length ξ – and reduces exactly to the classical Beer–Lambert law $e^{-s\sigma_m}$ in the limit $v \rightarrow 0$, by the check of Equation (34). For any $v > 0$, it exceeds the classical prediction at every finite s , consistent with Equation (19), with the size of the departure controlled jointly by how large the fluctuations are (v) and how far they extend (ξ).